



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

MP0: Setting Up a Kernel Development Environment

CS 423: Operating System Design

Fall 2026

Important Dates

- **MP0 is released today.**
- **The due date is 9/14 11:59:59 PM CT.**
- **No late submission, even 1 second late.**

Goals

- **You will configure your development environment for upcoming MPs.**
- **You will download, compile, and test your own kernel.**
- **The kernel source code will be a helpful reference.**

What you need

- **A Linux environment to compile your kernel and develop the MP.**
- **Another Linux environment to test your kernel/module.**

CS 423 Infra



CS 423 VM

- **For your development**
- **Provided by EngrIT**
- **You must use this VM to complete the MPs**
- **Ubuntu 24.04, No GUI**

CS 423 Qemu

- **Another VM for testing your kernel/module**
- **Q script creates an overlayfs on your VM, modification not saved**
- **Crash won't affect the host VM/your development, fast reboot**
- **... Yes, it is nested virtualization**

Connecting to your VM

- **Check your arrangement and power it ON first**
- **Use your favorite SSH client or tool like VSCode**
- **Install your public key (typing NetID password everytime?)**
- **We strongly recommend VSCode...**



Demo

Navigating your code/kernel source

- **Bootlin Elixir (Web)**

<https://elixir.bootlin.com/linux/>

- **VSCode + clangd (recommended)**

Needs some work to make it functional

```

64  /* for command line parsing */
65  static struct hstate * __initdata parsed_hstate;
66  static unsigned long __initdata default_hstate_max_huge_pages;
67  static bool __initdata parsed_valid_hugepagesz = true;
68  static bool __initdata parsed_default_hugepagesz;
69
70  /*
71   * Protects update
72   * free_huge_page
73   */
74  DEFINE_SPINLOCK(&hugetlb_lock);
75
76  /*
77   * Serializes fault
78   * prevent spurious
79   */
80  static int num_faults;
81  struct mutex *hugetlb_mutex;
82
83  /* Forward declaration
84  static int hugetlb_get_unmapped_area(
85  static void huge_page_fault(
86

```

Defined in 1 files as a variable:

mm/hugetlb.c, line 68 (as a variable)

Referenced in 1 files:

mm/hugetlb.c

- line 3652
- line 3815
- line 3846
- line 3860

```

194 module_init(test_module_init);
195
196
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200
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```

```

macro module_exit
provided by <linux/module.h>

#define module_exit(exitfn) \
    static inline exitcall_t __maybe_unused __exittest(void) { return exitfn; } \
    void cleanup_module(void) __copy(exitfn) __attribute__((alias(#exitfn))); \
    __CFI_ADDRESSABLE(cleanup_module, __exitdata);

// Expands to
static inline __attribute__((__gnu_inline__)) __attribute__((__unused__))
__attribute__((no_instrument_function)) exitcall_t __attribute__((__unused__))
__exittest(void) {
    return test_module_exit;
}
void cleanup_module(void) __attribute__((alias("test_module_exit")));
;

module_exit(test_module_exit);

```

Submission

- **Submit to Google Form (posted on the MP0 doc).**
- `dmesg | grep 'Linux version'`
- **Requires Illinois login.**
- **We will use your last submission for grading.**



Q/A